

Ch. 7 — P03 (K) Verify M/M/1 in SimPy

Problem Bank | Chapter 7 | Type: K | Difficulty:

Chapter: 7 Type: K Difficulty:

Objective

Implement an M/M/1 queue in SimPy and verify your simulation output against the analytical formulas from P01.

Setup

Use the same parameters as P01: $\lambda = 6/\text{hour}$, $\mu = 8/\text{hour}$.

Tasks

1. **Implement** a SimPy M/M/1 simulation function with the signature:
2. **Run** a single replication with `sim_time = 50_000` (time units = minutes). Compare your simulated W_q to the theoretical value from P01.
3. **Run 30 replications** and compute a 95% confidence interval for W_q . Does the CI contain the theoretical W_q ?
4. **Sweep** λ from 1 to 7.5 (keeping $\mu = 8$) in steps of 0.5. For each λ , run 10 replications and record mean W_q . Plot simulated W_q vs. $\rho = \lambda/\mu$, overlaying the theoretical curve. This is the “hockey stick” figure.
5. **Discuss:** At what ρ does the simulation become unreliable (high variance)? Why?

Submission

Submit a Jupyter notebook with:

- The SimPy model code (in a code cell or imported from `simdes`)
- The 30-replication CI result and comparison to theory
- The hockey-stick plot (publication quality, color-blind-safe)

- A written discussion (≤ 200 words)